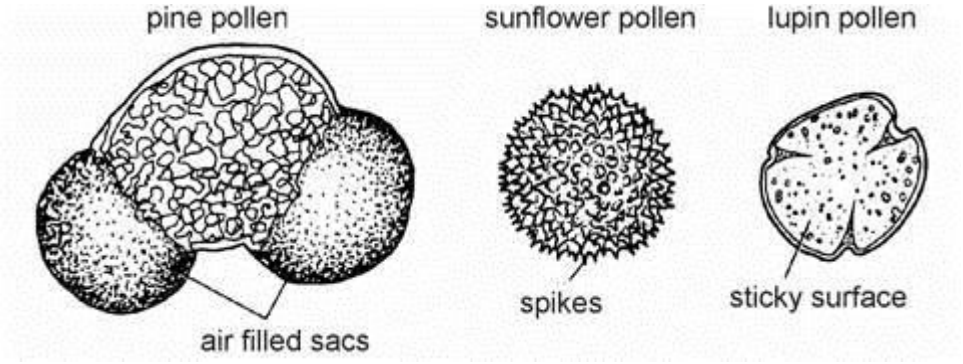
**Q1.** Most pollen grains are transferred from one flower to another either by **wind** or by **insects**.

Look at the drawings below which show pollen grains from three different plants.




(a) Using your observations:

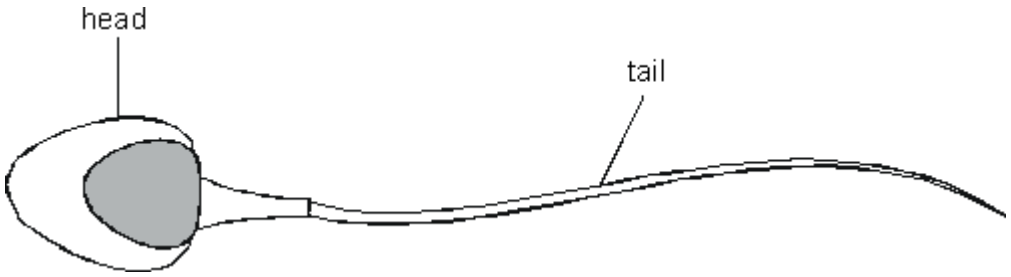
1. State the method by which each of these pollen grains is transferred.
2. Give a careful explanation for the method you have chosen each time.

Write your answers in the table.

| name of plant | method by which pollen is transfered | explanation for method chosen |
|---|---|---|
| pine | | |
| sunflower | | |
| lupin | | |

3 marks

The diagram shows a sperm cell.




(b) What is the **function** of each of the following structures in a sperm cell?

(i) Tail .....

.....

1 mark

(ii) Head .....

.....

1 mark

Maximum 5 marks

## Q1.

**Both the pollen transfer method and the explanation must be correct for the mark to be awarded, a reference to the feature only is insufficient.**

(a)

| plant | method | explanation | additional guidance | |
|---|---|---|---|---|
| pine | wind | large surface area **or** light | do **not** accept 'air sacs' | 1 |
| sunflower | insects | spikes get caught on insect **or** insect hair | do **not** accept 'spikes' | 1 |
| lupin | insects | sticks to insect | do **not** accept 'sticky surface' **or** 'insects like sticky things' **or** sticky things are sweet' | 1 |

(b) (i) to propel **or** move the sperm

*accept 'to swim'*

**1**

(ii) any **one** from

- to carry **or** contain **or** transfer genetic material

  *accept 'genes' **or** 'chromosomes'*
  ***or** 'nucleus **or** 'DNA' for genetic material*

- to penetrate the egg

  *accept 'to fertilise the egg'*

  *do **not** accept 'to control the cell'*

**1**

**[5]**